Worksheet — 1.3 Exchanging Data

Name: ___ ***Date:*** ___

OCR A Level Computer Science (H446) — Component 01, Section 1.3 (Compression, encryption & hashing; Databases; Networks; Web technologies)

Time: ~35 min.

1. Key-term match

Match each term (A–H) to its definition (1–8). Write the letter next to the number.

	Term		Definition
A	Lossy compression	1	Translates domain names into IP addresses
B	Hashing	2	A foreign key must reference a valid existing primary key
C	RLE	3	One-way function producing a fixed-length digest
D	DNS	4	Permanently removes data; original cannot be recovered
E	Referential integrity	5	Public + private key pair used for encryption
F	Asymmetric encryption	6	Replaces runs of repeated values with value + count
G	Primary key	7	Organising data to reduce redundancy & anomalies
H	Normalisation	8	Attribute(s) that uniquely identify each record

Answers: A- **B-** C- **D-** E- **F-** G- **H-**

2. Run Length Encoding (RLE)

(a) Encode the following pixel run as RLE pairs of *(value, count)*:

```
R R R R G G B B B W W W W W
```

Working space:

___

___

(b) Decode this RLE data back into the original sequence. Each pair is *(value, count)*:

```
(3,A) (1,B) (4,A) (2,C)
```

Working space:

___

(c) Give **one** type of data for which RLE would *increase* the file size, and say why.

____

____

3. Symmetric vs asymmetric encryption — fill in

Complete the blanks:

a) **Symmetric** encryption uses the __ key to encrypt and decrypt. Its main weakness is the ____ problem.

b) **Asymmetric** encryption uses a __ key (shared openly) and a __ key (kept secret).

c) Data encrypted with someone's **public** key can only be decrypted with their __ key.

d) Asymmetric is __ (faster / slower) than symmetric.

e) HTTPS uses asymmetric encryption to securely exchange a __ session key, then uses __ encryption for the bulk data because it is faster.

4. Normalisation

An unnormalised `STUDENT_COURSES` table is shown below.

StudentID	StudentName	Courses	TutorID	TutorName
S01	Amy Khan	Maths, Physics	T1	Mr Bell
S02	Ben Cole	Maths	T1	Mr Bell
S03	Cara Lee	Physics, Chemistry	T2	Ms Frost

(a) State **one** repeating group that breaks **1NF**, and explain how to fix it.

____

____

(b) The `TutorName` depends on `TutorID`, not on the student. What type of dependency is this, and which normal form does it break?

____

(c) Redesign the data into **3NF**. Sketch the tables (table name + columns). Underline the **primary key** in each and mark any **foreign key** with (FK).

Table 1: ____________________ (________)

Table 2: ____________________ (________)

Table 3: ____________________ (________)

Table 4 (if needed): ____________________ (________)

5. Write the SQL

Use these tables: `Student(StudentID, Name, TutorID, YearGroup)` and `Tutor(TutorID, TutorName)`.

(a) Return the **Name** and **YearGroup** of every student in Year 12, sorted by Name (A–Z).

(b) Return each student's **Name** alongside their **TutorName** (join the two tables on TutorID).

(c) Change the **YearGroup** to 13 for the student whose StudentID is `S01`.

6. TCP/IP layer matching

Write each protocol in the row of the layer it belongs to.

Protocols: `HTTP`, `TCP`, `IP`, `Ethernet`, `SMTP`, `UDP`, `Wi-Fi`, `FTP`

TCP/IP layer	Protocol(s)
Application	___________
Transport	___________
Network / Internet	___________
Link / Data link	___________

7. Short answer

(a) In **packet switching**, give two things that are added to each packet so it can travel the network and be reassembled.

(b) What does **DNS** do, and why is caching used?

(c) In **PageRank**, why is a link from a high-ranked page worth more than a link from a low-ranked page?

(d) Give one reason why login/payment checks must run **server-side** rather than only client-side.

🧠 Challenge box

A website stores user passwords. Explain why the passwords should be **hashed** rather than **encrypted**, and describe what happens when a user later logs in (refer to *one-way*, *deterministic*, and *collision*).